

MRS and MRSC File Format Specification

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1 Overview

The roll scanner of Bern University of Applied Sciences (BFH) records a roll with two line cameras, a monochrome camera in transmitted light and an RGB camera in reflected light. An encoder wheel on the paper triggers both cameras every 0.2 mm (Debrunner 2011, p. 39). The data of the two cameras are written to `*.mrs` (transmitted light) and `*.mrsc` (reflected light) (Debrunner 2011, p. 51). A reference implementation is `mrs2image.py` in <https://github.com/pfefferniels/rollscan2image>.

Offsets and lengths are in bytes. Numbers in header fields are decimal ASCII, zero-padded to ten digits.

2 Container

Both formats consist of a 124-byte header, raster data in chunks, each preceded by a marker, and an end marker. MRS files end with a metadata trailer.

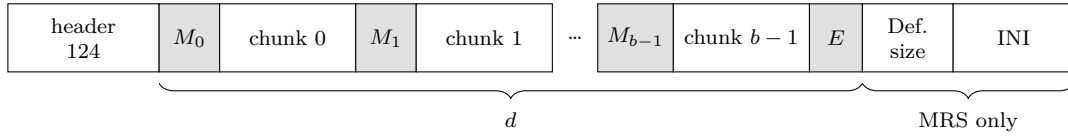


Figure 1: Container layout, not to scale. M_i is the marker of chunk i , E the end marker.

Offset	Length	Content
0	30	Headsize in Bytes = h
30	64	$\sqcup$ Date = $DD.MM.YYYY$, MRS kk Nr = $V.x.y$, Scan Number = s ;
94	30	Datasize in Bytes = d
124	10	marker \$000000000
134	kL	chunk 0
	10	marker \$000000001
	kL	chunk 1, and so on
$114 + d$	10	end marker \$999999999
$124 + d$	30	MRS only: Def.size in Bytes = m
$154 + d$	m	MRS only: metadata

Table 1: Byte layout. $\sqcup$ is a space.

- The header fields follow each other without separators.
- h is the length of the middle field, always 64.
- kk is SW in MRS and Co in MRSC, $x.y$ is the format version and s the scan number.
- d is the number of bytes from offset 124 to the end of the end marker.
- A marker is \$ followed by the chunk number in nine digits, starting at 0.

- A chunk holds k lines of L bytes each (Table 2).

The number of chunks b and the number of lines z follow from d :

$$b = \left\lceil \frac{d - 10}{kL + 10} \right\rceil \quad z = \frac{d - 10(b + 1)}{L}$$

3 Rasters

	MRS	MRSC
Camera	transmitted light, monochrome	reflected light, RGB
kk	SW	Co
Pixels per line n	2048	2098
Bytes per pixel	1 (grey)	3 (red, green, blue)
Bytes per line L	2048	6294
Lines per chunk k	1000	1
Line spacing	0.2 mm	0.2 mm
Pixel spacing, nominal	0.22 mm	0.21 mm
Holes	bright	colour of the backing
Metadata trailer	yes	no

Table 2: The two rasters. Nominal pixel spacing after (Debrunner 2011, p. 40).

Values are unsigned 8-bit integers. Line 0 is the start of the scan. A line spans the camera’s field of view, which extends beyond both paper edges. Both cameras scan across the roll in the same direction. The two rasters are not registered to each other. The pixel spacing across the roll is not stored. It is determined per scan from **Roll Width** or from the track grid of the roll.

4 Metadata trailer

After the end marker, MRS files contain the 30-byte field **Def.size in Bytes** = m , followed by m bytes of INI text up to the end of the file. Lines have the form *key=value* and are grouped under section names in square brackets. Line endings are CR LF, the encoding is Latin-1. Unset values are empty.

The keys are listed in Table 3. N is the value of **Number Of Tracks**. **Roll Width**, **First Track** and **Last Track** are in mm, the track positions measured from the same paper edge. **Play Speed** is in mm/s.

Section	Keys
[Soundtrack 1]	Soundtrack Name, Composer First Name, Composer Last Name, Arranger First Name, Arranger Last Name, Interpret First Name, Interpret Last Name
[Rolltype]	Name, Track Shifts, Control Track Position, Number Of Tracks, Roll Width, First Track, Last Track, Track Alignement, Track Width, Trackgap, Min Track Length, Track 1 ... Track N
[Definition]	Title, Recording Year, Recording Location, Producer, Running Time, Play Speed, Notes, Catalogue Number, Catalogue Number 2, Is Replica, Paper Type, Roll Length, Box State

Table 3: Sections and keys of the metadata trailer, in file order.

5 Settings file

The recording software writes a CSV file with the base name of the MRSC file. Every field is terminated by a semicolon. The first line is `DateTime;Group;Variable;Value;Unit;. Group Images.settings` holds settings without unit, group `Images` holds measurements with the unit in square brackets.

Variable	Content
MRSC_Brightness_Correction_Enable	brightness correction (True, False)
MRSC_Brightness_Courve_A, _B, _C	coefficients A , B , C of the brightness curve
MRSC_SpatialCorrOfOneColor	colour row offset δ in lines
MRSC_DetectPencil j _Enable	pencil detection in range j (True, False)
MRSC_DetectPencil j _minValues	lower colour of range j
MRSC_DetectPencil j _maxValues	upper colour of range j
MRSC_DetectTape j _Enable	tape detection in range j (True, False)
MRSC_DetectTape j _minValues	lower colour of range j
MRSC_DetectTape j _maxValues	upper colour of range j
MRSC_DetectedPixels_Overwrite	overwrite detected pixels (True, False)
MRSC_Length	length of the scan in m
MRSC_PensilPixels	pencil pixels in permille
MRSC_TapePixels	tape pixels in permille

Table 4: Variables of the settings file, $j \in \{1, 2\}$. Colours are written as Color [A= a , R= r , G= g , B= b].

6 Colour image

The PNG delivered by the scanner is computed from the MRSC raster by colour registration, brightness correction and mirroring. Scan lines become columns, pixels become rows:

$$p(x, n - 1 - s, i) = \min \left(255, \left\lfloor r(x + \delta_i, s, i) \cdot \frac{c_{\max}}{c(s)} \right\rfloor \right)$$

- $r(x, s, i)$ is the MRSC value in line x , pixel s and channel $i \in \{R, G, B\}$.

- $\delta_R = 2\delta$, $\delta_G = \delta$, $\delta_B = 0$, with $\delta = \text{MRSC_SpatialCorrOfOneColor}$.
- $c(s) = As^2 + Bs + C$, and $c_{\max}$ is the maximum of $c(s)$ for $0 \leq s < n$.
- If `MRSC_Brightness_Correction_Enable` is `False`, $c_{\max}/c(s) = 1$.
- The image has $z - 2\delta$ columns and n rows.

References

Debrunner, Daniel. 2011. “Die Entwicklung des Musikrollenscanners der Berner Fachhochschule – aus Musikrollenbildern wird Musik – die elektronische Steuerung der Welte-Philharmonie-Orgel.” In *Wie von Geisterhand. Aus Seewen in die Welt. 100 Jahre Welte-Philharmonie-Orgel*, 35–63. Seewen.